Table 1: Data and locked transfer protocol.

Quantity	Value
Syndrome bits b	16
Source rows	75,598
Represented shots	70,000,000
Development rates	indices 2–4
Confirmation rates	0.005, 0.010
Confirmation shots	20,000,000
Selected history width	2 cohorts

FAST HAMMING PRODUCT-KERNEL TRANSFER FOR SURFACE-CODE SYNDROME PROBABILITIES

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Abstract. Transferring decoder probabilities across physical fault rates requires a representation that respects the binary geometry of syndrome histories. An equal-class-prior Hamming product kernel is studied on the complete public distance-three archive associated with a cryogenic decoder. The kernel table is applied by one two-point transform per syndrome bit, reducing a dense $2^b \times 2^b$ kernel multiplication to $O(b2^b)$ arithmetic and $O(2^b)$ storage. A normalization theorem bounds probability error from class-score errors. History width and kernel bandwidth are selected by rolling rate holdout; the two highest physical fault rates, comprising twenty million simulated shots, remain untouched. On this confirmation set the product kernel has log loss 0.2871, versus 0.5358 for a hierarchical blend, 0.6333 for tuned logistic regression, and 0.6931 for a constant predictor. Exact lookup has higher balanced accuracy, 0.9189 versus 0.9038, but substantially worse log loss, 0.7724. Thus the supported improvement concerns calibrated probability transfer rather than every classification metric. A separate hierarchical operator audit confirms decreasing storage and approximation error. Optimized dense multiplication is faster through 1024 records, whereas the hierarchy crosses over at 2048 records on the recorded platform. The contribution is the exact binary-cube kernel algorithm and its authenticated transfer evaluation, not a hardware decoder or a hierarchical speedup.

1. Authenticated transfer task. The source is Zenodo record 11166209, version 1.0.1, under CC BY 4.0 [1]. Seven CSV files summarize ten million Stim-generated shots each at increasing physical fault rates. Every row contains a binary 4×4 syndrome history, a logical label, and its shot multiplicity. The archive has 75,598 rows representing 70 million shots. It is an authenticated simulation record, not quantum-hardware data.

Rates indexed 0–4 form training and rolling development cohorts. Rates 5 and 6, with physical fault rates 0.005 and 0.010, form the fixed confirmation set. All metrics use equal total weight for the two logical classes while retaining within-class shot multiplicities. The model therefore estimates an equal-class-prior conditional probability, matching balanced accuracy rather than the archive class prevalence. Table 1 records the resulting source and holdout contract.

2. Binary-cube product kernel. Encode a syndrome as $s \in \{0,1\}^b$. For bandwidth $0 \le \rho \le 1$, set

$$K_\rho(s,t) = \rho^{d_H(s,t)}. \tag{2.1}$$

For class c, let $q_c(t)$ be normalized weighted counts in the selected history window. Scores and probabilities are

$$a_c(s) = \sum_t K_\rho(s,t) q_c(t), \qquad p(s) = \frac{a_1(s)}{a_0(s) + a_1(s)}. \tag{2.2}$$

The registered grid contains history widths 1–4 and logarithmically spaced bandwidths including $\rho = 0$. Each development rate is predicted only from lower-rate cohorts. The selected pair is refitted on the two cohorts ending at rate index four.

Table 2: Class-balanced confirmation metrics over the two held-out rates.

Method	Log loss	Brier	Bal. acc.
Constant	.6931	.2500	.5000
Logistic	.6333	.1299	.8412
Exact lookup	.7724	**.0715**	**.9189**
Hamming kernel	**.2871**	.0767	.9038
Hierarchy blend	.5358	.1294	.8410

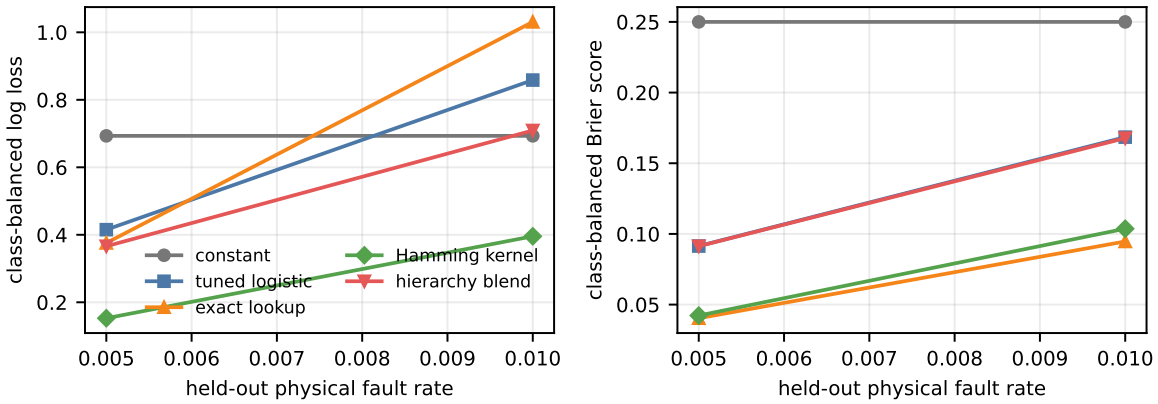


Fig. 1: Rate-resolved log loss and Brier score on both confirmation cohorts. The product kernel retains the log-loss advantage as the fault rate doubles.

Theorem 1 (Tensor factorization and transform cost)**.** *In lexicographic binary order,*

$$K_\rho = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}^{\otimes b}. \tag{2.3}$$

Both class-score tables can be computed in $2b2^b$ *pair updates, using* $O(2^b)$ *working storage per class, without forming* K_ρ*.*

Proof. The Hamming distance is additive over coordinates, hence $\rho^{d_H(s,t)} = \prod_{j=1}^{b} \rho^{\mathbf{1}\{s_j \neq t_j\}}$. The factor for bit j equals the displayed two-by-two matrix entry, proving the Kronecker identity. Applying one factor updates every disjoint pair of table entries, requiring 2^b scalar-output updates per bit and class. In-place pair updates store only the current score table. □

Theorem 2 (Probability perturbation)**.** *Let* $a_0, a_1 \geq 0$*,* $d = a_0 + a_1 > 0$*, and suppose* $|\widetilde{a}_c - a_c| \leq \varepsilon_c$ *with* $\varepsilon_0 + \varepsilon_1 < d$*. Then*

$$\left| \frac{\widetilde{a}_1}{\widetilde{a}_0 + \widetilde{a}_1} - \frac{a_1}{d} \right| \leq \frac{a_0\varepsilon_1 + a_1\varepsilon_0}{d(d - \varepsilon_0 - \varepsilon_1)}. \tag{2.4}$$

Proof. Writing $\delta_c = \widetilde{a}_c - a_c$ and clearing denominators gives

$$\widetilde{p} - p = \frac{a_0\delta_1 - a_1\delta_0}{d(d + \delta_0 + \delta_1)}. \tag{2.5}$$

The numerator is at most $a_0\varepsilon_1 + a_1\varepsilon_0$ in magnitude, and the second denominator factor is at least $d - \varepsilon_0 - \varepsilon_1$. □

3. Confirmation results. Table 2 separates probability calibration from thresholded classification. The Hamming kernel has the lowest log loss. Exact lookup has the highest balanced accuracy and lowest Brier score, but its near-zero probabilities are severely penalized when an unseen-rate label contradicts the lookup. The transform therefore improves the primary probabilistic metric without dominating every metric.

At fault rate 0.005, kernel log loss is 0.1527 and balanced accuracy is 0.9464. At 0.010 the values are 0.3955 and 0.8707. Figure 1 shows the degradation directly rather than merging the two regimes into one favorable number.

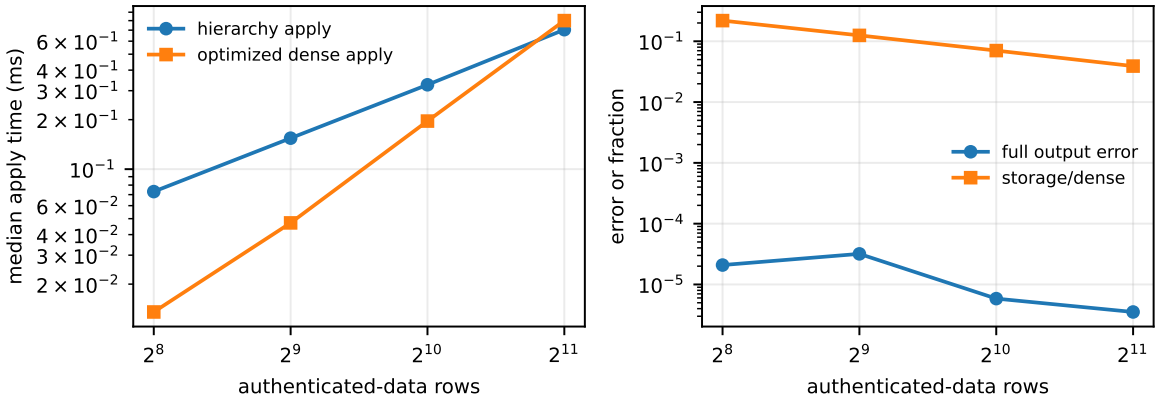


Fig. 2: Measured hierarchy and dense-apply time, together with approximation error and storage fraction. The compressed representation saves storage and crosses the dense timing only at the largest tested size.

4. Representation audit. A hierarchical approximation is not needed by the $b = 16$ product kernel: the exact tensor transform already has linear-in-b table complexity. The prior hierarchical route is retained as a negative control. On authenticated subsets of 256–2048 rows, its relative output error decreases from 2.09×10^{-5} to 3.53×10^{-6} and stored fraction decreases from 0.219 to 0.039. Optimized dense application is faster through 1024 records, whereas the hierarchy is faster at 2048 records on the recorded platform; Fig. 2 reports the complete crossover without extrapolation.

5. Scope. The experiment tests rate transfer on simulated distance-three syndromes. It does not measure threshold behavior, hardware drift, decoding latency, or larger code distances. Exact lookup remains preferable when thresholded accuracy alone determines the objective. The product-kernel contribution is narrower: rolling-selected Hamming smoothing supplies substantially better held-out probability calibration, an exact $O(b2^b)$ transform, and a conditioning bound for approximate class scores. All rate deterioration and the negative hierarchical timing result remain part of the evidence.

References.

[1] Victor Yon and Gebremedhin A. Dagnew. A memristive neural decoder for cryogenic fault-tolerant quantum error correction—syndromes dataset, 2024.